of LCD1 and pins PC0 through PC2 for control pins of LCD1. Temperature range can be defined in the source code, within which LCD1 displays the message 'it's normal temperature' and green LEDs (LED5 and LED8) glow. If temperature crosses the maximum of the range, the message on the display changes to 'it's too hot' and red LEDs (LED3 and LED6) glow. In case temperature goes below the minimum limit, the message changes to 'it's too cold' and blue LEDs (LED4 and LED7) glow.

The triggering value for light intensity can also be defined in the source code. If the intensity of light goes below the predefined value, the relay connected through CON2 is driven by port pin PD7 of IC3 to switch on the bulb and the LCD shows the message 'bulb on.'

The power to the whole system is provided using a 12V adaptor. Regulator IC1 provides a 5V output, which is further regulated by IC2 to 3.3 V.

## Software

The program is written in 'C' language. It is compiled and programmed into the target device using WinAVR. The html page embedded in the source code can be accessed from a remote place to find out the readings of the temperature and light sensors. The section of the program defining the html page is shown in Fig. 5.

To access the system, you have to change the mac address and IP address corresponding to your network arrangement in the line mentioned below in the source code:

```
static uint8_t mymac[6] = {0x54,0x90,
0x58,0x10,0x65,0x77};
```

```
static uint8_t myip[4] =
{192,168,2,248};
```

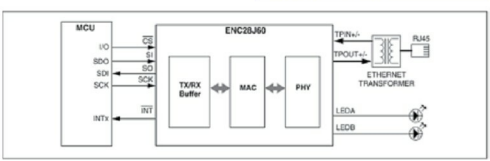



*Fig. 4: Simplified connection diagram*

### PARTS LIST

*Semiconductors:*

| | |
|---|---|
| IC1 | - 7805, 5V regulator |
| IC2 | - LM1117, 3.3V LDO |
| IC3 | - ATmega128 microcontroller |
| IC4 | - ENC28J60 Ethernet controller |
| IC5 | - TMP275 temperature sensor |
| IC6 | - APDS-9300 light photo sensor |
| LED1 | - 3mm green LED |
| LED2-LED8 | - 3mm red LED |
| D1-D2 | - 1N4007 rectifier diode |
| T1 | - BC547 npn transistor |

*Resistors (all ¼-watt carbon):*

| | |
|---|---|
| R1-R8 | - 220-ohm |
| R9-R14 | - 10-kilo-ohm |
| R15-R18 | - 50-ohm |
| R19 | - 2-kilo-ohm |
| VR1 | - 10-kilo-ohm preset |

*Capacitors:*

| | |
|---|---|
| C1 | - 10µF, 16V electrolytic |
| C2 | - 1µF, 16V electrolytic |
| C3-C8 | - 22pF ceramic |
| C9-C13 | - 104pF ceramic |

*Miscellaneous:*

| | |
|---|---|
| LCD1 | - 16×2 LCD |
| S1 | - Push-to-on switch |
| JACK | - 12V DC jack |
| CON1 | - 2-pin connector for 5V, 1C/O relay |
| CON2 | - Ethernet female connector (RJ45) |
| $X_{TAL1}$ | - 7.37MHz crystal |
| $X_{TAL2}$ | - 32.76MHz crystal |
| $X_{TAL3}$ | - 25MHz crystal |

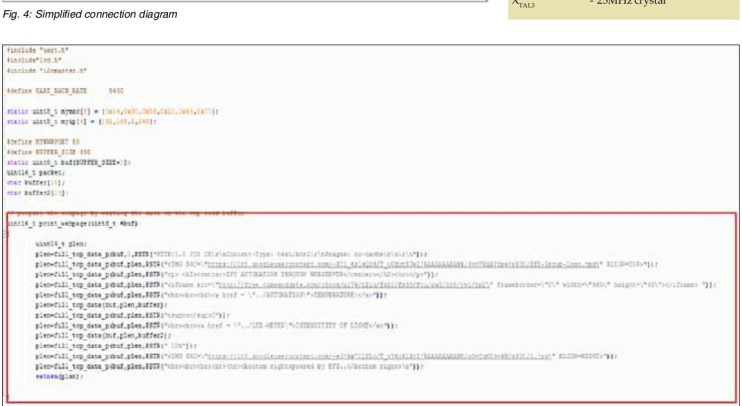


*Fig. 5: html page in the source code*